

Dr. Soumyadip Bandyopadhyay

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Current address	Dr. Soumyadip Bandyopadhyay D 66 BITS Pilani K K Birla Goa Campus NH 17B Bypass Road, Zuarinagar Goa-403726 Country: India	
Education	Indian Institute of Technology, Kharagpur, West Bengal, India Degree: <i>Ph.D.</i> • Computer Science and Engineering • Formal Verification Research Group • Professor Chittaranjan Mandal and Professor Dipankar Sarkar¹ • Thesis Title: Path Based Equivalence Checking of Petri Net Representation of Programs for Translation Validation • Thesis Submission: September 2015 • Thesis Defended: December 2016 • Formally Awarded: August 2017 West Bengal University of Technology, Kolkata, , West Bengal, India Degree: <i>B.Tech</i> • Computer Science and Engineering • C.G.P.A: 8.43 in scale of 10.0	2009–2016 2004-2008
Work Experience	• Research Scientist, ABB Corporate Research (July 2023 to Till date) • Senior Formal Verification Engineer, NVIDIA, (May 2022 to June 2023) • Assistant Professor, Dept of CSIS, BITS Pilani K K Birla Goa Campus (December 2018 to May 2022) • Post Doctoral Fellow, System Analysis and Modeling Group, Hasso Plattner Institute, Potsdam, Germany (August 2017 to October 2018) • Assistant Professor, Dept of CSIS, BITS Pilani K K Birla Goa Campus (December 2016 to July 2017) • Visiting Faculty, Dept of CSIS, BITS Pilani K K Birla Goa Campus (September 2015 to December 2016) • Junior Project Assistant, Advanced VLSI Lab, IIT Kharagpur (August 2008 to December 2012)	
Research interest	• Generative AI • Formal methods • Program Equivalence • Software Verification • Process automation • PLC verification • Data path verification • High Level Synthesis • Model driven engineering	
Honours and Awards	• Finalist at the Global HackSummit 2024 hosted by ABB • Best paper ICSoft 2021 • Awarded Bronze medal in ACM SRC Grand final Event 2021 as a Mentor. • Selected in 7th HLF as top 50 young researcher in computer science • Humboldt-Somed HPI Post Doctoral fellowship Germany, 2017 • Tata Consultancy Service Innovation Lab Research Fellowship, 2012 • Academic of science Czech Republic Scholarship, 2007	

¹Dept. of Computer Science and Engineering

Professional activities	• Reviewer at CAV-2014, EMSOFT-2015, DAC 2020, ACM TOSEAM, Acta Informatica, • PC member ICSOFT 2018,2019, 2020, 2021, 2022, 2023, VLSI D 2023, INDICON, 2023, MPM4CPS 2021, 2022, 2023, ISEC 2018, 2019, 2020, 2021, 2022 • Senior IEEE Member, ACM Member, INSTICC Member • PERR 2022 (Co-chair) co-located with FLOC 2022
Journals Publications	• Soumyadip Badyopadhyay, Dipankar Sarkar, Chittaranjan Mandal, Holger Giese, "Translation Validation of Coloured Petri Net Models of Programs on Integers", Acta Informatica • Soumyadip Badyopadhyay, Dipankar Sarkar, Chittaranjan Mandal, "Equivalence checking of Petri net models of programs using static and dynamic cut-points", Acta Informatica • Soumyadip Bandyopadhyay, Dipankar Sarkar, Kunal Banerjee, Chittaranjan A. Mandal, Krishnam Raju, "A Path Construction Algorithm for Translation Validation using PRES+ Models", Parallel processing letters
Conference and workshop Publications	• Soumyadip Bandyopadhyay and Santonu Sarkar, "Antarbhukti: Verifying Correctness of PLC Software during System Evolution", ATVA 2025 • Soumyadip Bandyopadhyay and Raoul Jetley, "Pn4PLC: Verification of Software Upgrade for PLC Code", FSE Companion 2025 • Md Tauseef Alam, Sorbajit Goswami, Khushi Singh, Raju Halder, Abyayananda Maiti, Soumyadip Bandyopadhyay, "SolGen: Secure Smart Contract Code Generation Using Large Language Models Via Masked Prompting", ISEC 2025 • Heiko Koziol, Virendra Ashiwal, Soumyadip Bandyopadhyay, Chandrika K R, "Automated Control Logic Test Case Generation using Large Language Models", ETFA 2024 • Rakshit Mittal, Dominique Blouin, Anish Bhoje and Soumyadip Bandyopadhyay, "Solving the Instance Model-View Update Problem in AADL", MODELS 2022 • Rakshit Mittal, Dominique Blouin, Soumyadip Bandyopadhyay, "PNPEq: Verification of Scheduled Conditional Behavior in Embedded Software", APSEC 2021 • Rakshit Mittal, Rochishnu Banerjee, Dominique Blouin, Soumyadip Bandyopadhyay, "Towards an Approach for Translation Validation of Thread-level Parallelizing Transformations using Colored Petri Nets", ICSOFT 2021 (Best Paper) • Rakshit Mittal, Rochishnu Banerjee, Santonu Sarkar, Soumyadip Bandyopadhyay, "Translation Validation of Loop involving Code Optimizing Transformations using Petri Net based Models of Programs", Petri Nets workshop 2020 • Shivam, Nilanjana Goswami, Veeky Baths, Soumyadip Bandyopadhyay, "AES: Automated Evaluation Systems for Computer Programming Course", ICSOFT 2019 • Soumyadip Bandyopadhyay, Dipankar Sarkar, Chittaranjan Mandal, "SamaTulyataOne: A Path Based Equivalence Checker", ISEC 2019 • Santonu Sarkar, Prateek Kandelwal, Soumyadip Bandyopadhyay, Holger Giese, "Analysis of GPGPU Programs for Data-race and Barrier Divergence", ICSOFT 2018 • Soumyadip Bandyopadhyay, Santonu Sarkar, Dipankar Sarkar and Chittaranjan Mandal; SamaTulyata, "An Efficient Path Based Equivalence Checking Tool", ATVA 2017 • Soumyadip Bandyopadhyay, Santonu Sarkar and Kunal Banerjee, "An End-to-End Formal Verifier for Parallel Programs", ICSOFT 2017 • Soumyadip Bandyopadhyay and Kunal Banerjee, "PRESGen: A Fully Automatic Equivalence Checker for Validating Optimizing and Parallelizing Transformations", HPDC workshop 17 • Soumyadip Bandyopadhyay, Dipankar Sarkar and Chittaranjan Mandal, "An efficient path based equivalence checking for Petri net based models of programs", ISEC-2016 • Soumyadip Bandyopadhyay and Kunal Banerjee, "Implementing an Efficient Path Based Equivalence Checker for Parallel Programs", HPDC workshop 16 • Kunal Banerjee, Soumyadip Bandyopadhyay, and Santonu Sarkar, "Data-Race Detection: The Missing Piece for an End-to-End Semantic Equivalence Checker for Parallelizing Transformations of Array-Intensive Programs", PLDI workshop 2016 • Soumyadip Bandyopadhyay, Dipankar Sarkar and Chittaranjan Mandal, "Validating SPARK: High Level Synthesis compiler", ISVLSI-2015 • Soumyadip Bandyopadhyay, Dipankar Sarkar and Chittaranjan Mandal, "A Path-Based Equivalence Checking Method for Petri net based Models of Programs", ICSOFT-EA-2015 • Soumyadip Bandyopadhyay, Dipankar Sarkar, Chittaranjan A. Mandal, "An Efficient Equivalence Checking Method for Petri net based Models of Programs", ICSE 2015 • Soumyadip Bandyopadhyay, Kunal Banerjee, Dipankar Sarkar, Chittaranjan A. Mandal, "Translation Validation for PRES+ Models of Parallel Behaviours via an FSM-D Equivalence Checker", VDAT 2012
Poster Publications	

	• Soumyadip Bandyopadhyay, "Behavioural verification using Petri net based models of programs", POPL-2015 (ACM student research competition) • Soumyadip Bandyopadhyay, Dipankar Sarkar, Chittaranjan A. Mandal, "Translation Validation using Path-Based Equivalence Checking of Petri net based Models of Programs", WEPL 2015
Book chapter	• Bedir Tekinerdogan, Rakshit Mittal, Rima Al-Ali, Mauro Iaconod, Eva Navarroe, Soumyadip Bandyopadhyay, Ken Vanherpen and Ankica Barisic, "A feature-based ontology for cyber physical systems", Chapter 3, Book Title : Multi-Paradigm Modelling Approaches for Cyber-Physical, Elsevier Press. ISBN No. 9780128191064 • Holger Giese , Dominique Blouin , Rima Al-Ali , Hana Mkaoua, Soumyadip Bandyopadhyay, Mauro Iacono, Moussa Amrani, Stefan Klikovits and Ferhat Erata "An ontology for multiparadigm modelling", Chapter 4, Book Title: Multi-Paradigm Modelling Approaches for Cyber-Physical, Elsevier Press. ISBN No. 9780128191064 • Dominique Blouin, Rima Al-Ali, Holger Giese, Stefan Klikovits, Soumyadip Bandyopadhyay, Ankica Barisic and Ferhat Erata, "An integrated ontology for multi-paradigm modelling for cyberphysical systems" , Chapter 5, Book Title: Multi-Paradigm Modelling Approaches for Cyber-Physical, Elsevier Press, ISBN No. 9780128191064
Patent	• Soumyadip Bandyopadhyay and Raoul Jetley, "LLM-driven Accelerated Code Generation" (Approved by business) • Avijit Mandal, Mrutyunjay G, Raoul Jetley, Soumyadip Bandyopadhyay "Generative AI based multi-language Support" (Approved by business)
Under preparation	• Soumyadip Bandyopadhyay and Santonu Sarkar, " Trustworthy LLM-driven Code Migration for Industrial Control Software " • Soumyadip Bandyopadhyay and Santonu Sarkar, " Trustworthy LLM-driven Evolution of Industrial Control Software "
Sponsored and Consultancy Projects	• "APP based learning for Python program" Funding Agency: 6th Sense and AGH advisor Duration : 2020-2022, Amount: 15.81L • "Modelling and Verification of Bio-Inspired system", Funding Agency: DST under BIO-CPS incubation BITS Pilani, Duration 2021-2022, Amount: 20L • "AES: Automated Evaluation Systems for Computer Programing Course in Any University" Funding Agency: BITS Pilani, Duration: 2018-2021, Project Amount : 2L • "SamaTulyata: Automated Evaluation for Computer Programming Course " Funding Agency: TLC BITS Goa, Duration: 2019-2020, Project Amount: 1L
Industrial Projects	• Execution level hardware formal verification for AI Supercomputer. • Explore the potential of Generative AI and Large Language Model (LLM) to support engineering • Compiler equivalence between OLD control compiler vs NGT compiler. • Proving the modern system to be functionally equivalent to the Heritage systems. Building a tool for equivalence checking between modern systems vs heritage systems. • Verification of the drift detection model • Hallucination detection of LLM-generated control code using the model-checking approach.
Teaching at BITS Goa	• Compiler Construction (WILP Spring 2019, Spring 2020) • Theory of Computation (Autumn 2019) • Parallel computing (Spring 2022) • Data Structure and Algorithm (Spring 2022) • Computer Architecture (WILP Autumn 2020) • Object-oriented programming Lab (Autumn 2020) • Principal of programming language (Autumn 2021) • Computer Programming Lab (Spring 2021)
Teaching at NVIDIA and ABB	• Road map of Functional verification for ML-based hardware accilarator • Software engineering for AI
Tool develop	• Antarbhukti for software evolve • ABB Co-pilot for PLC code generation and unit testing • SamaTulyata: Eclipse based Program equivalence tool using Petri net • Autoval: Automatic evaluation of student's Program using program equivalence • Raise-2: Test case based analysis tool for Computing course

- **CatGrader:** Category based grading for computer programming course.
- **OsarteDM:** Model transformation Tool for Cyber physical system

Programming Python, C, Verilog, Linux Shell scripting, L^AT_EX, C++.

GenAI Model Azure OpenAI, LLaMA, Gemini, Claude

Formal Tools worked with KLEE, CBMC, JasperGold, VC Formal, Helena, SAL, Pluto, Par4All, CPN, OpenAI

Language known English and Bengali

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